

$$\begin{aligned}
 M_{t_1,qs} &= K_{R,t_1} t_1^{\top} (e_R - e_{R,0}) \\
 &= (15 \text{ N.m/rad}) \cdot \left(6.57 \cdot \frac{\pi}{180} \text{ rad} \right) \approx 1.72 \text{ N.m}
 \end{aligned}$$